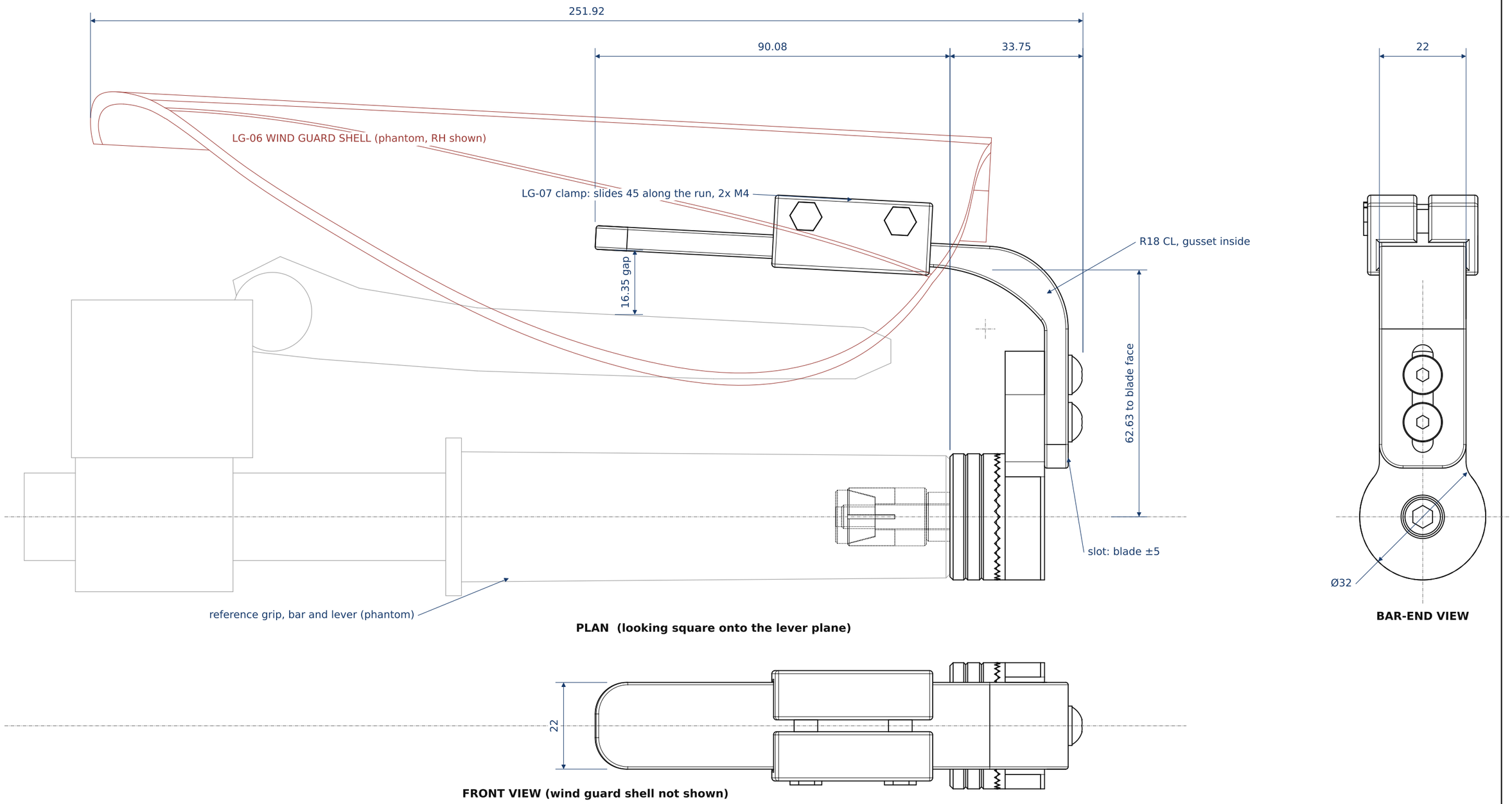



**PLAN (looking square onto the lever plane)**

**BAR-END VIEW**




**FRONT VIEW (wind guard shell not shown)**

**NOTES**

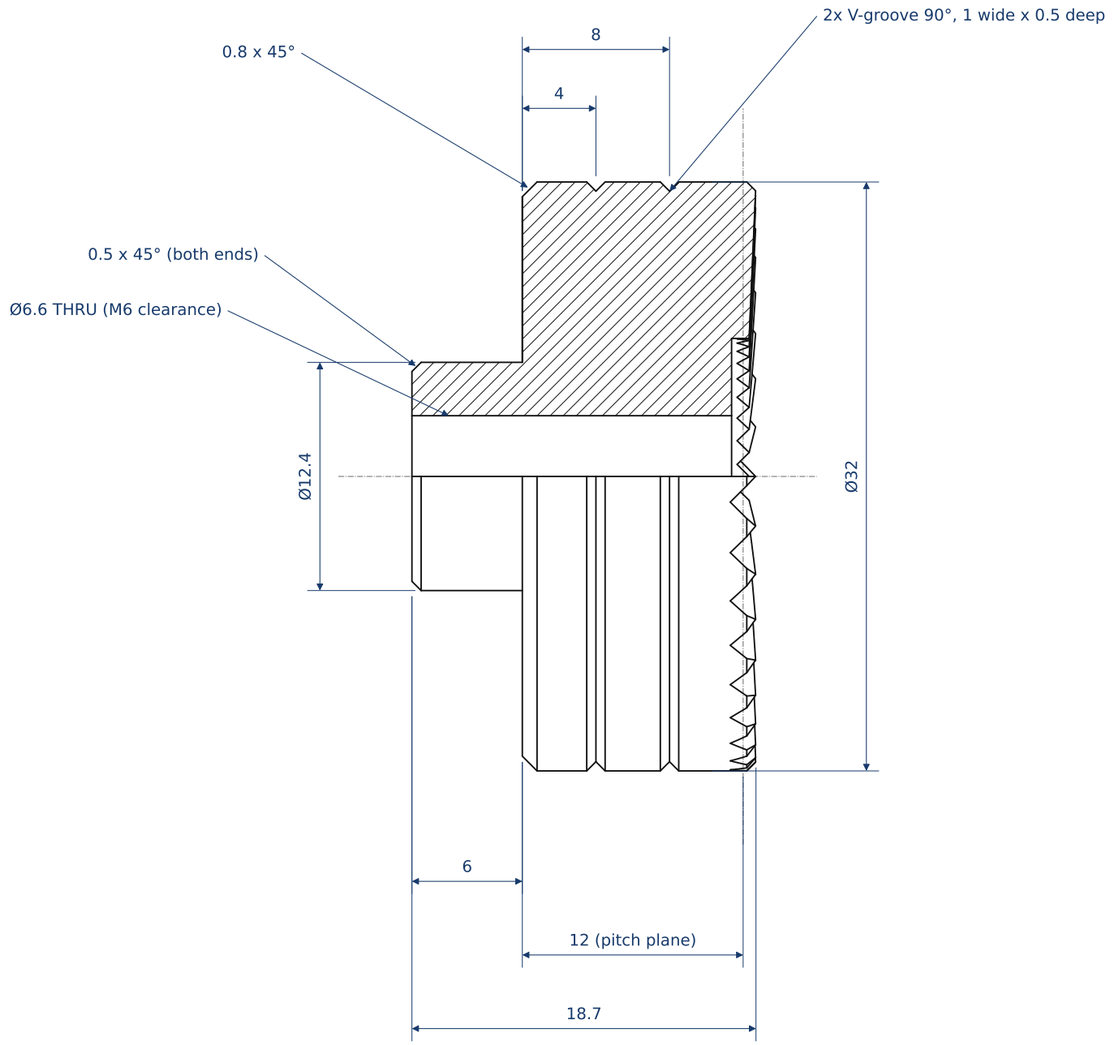
1. Drawing plane = lever plane. Turn the arm on its 36-tooth serration (10° steps) to put the blade on the lever line.
2. Clearances shown for the reference sport-bike lever; keep ≥ 10 mm all round, re-check at full lever travel.
3. Mass per side ≈ 123 g, 218 g with the wind guard.
4. Left side: same parts, arm turned over (they are symmetric about the lever plane), except the wind guard shell: LG-06 is made as a mirror-image pair.
5. Wind guard shell drawn as a phantom in the plan; its profile and fixing are on sheet 6.

RACING LEVER GUARD / UNIVERSAL BAR-END FIT

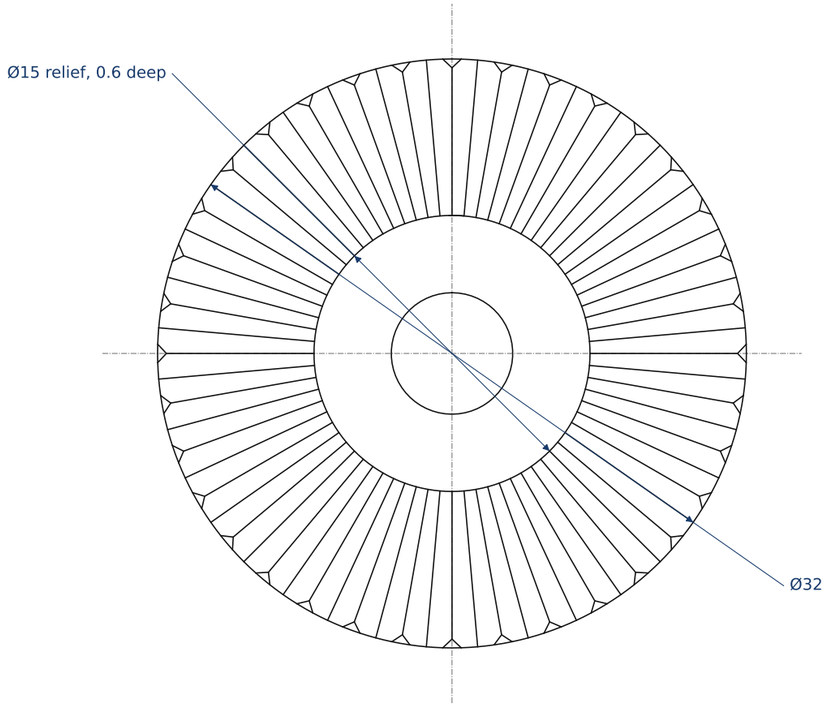
# GENERAL ARRANGEMENT

**LG-00**

| MATERIAL | | | | FINISH | |
|---|---|---|---|---|---|
| see part sheets | | | | see part sheets | |
| SCALE | UNITS | QTY / SIDE | GENERAL TOLERANCES | PROJECTION | SHEET |
| 1 : 1 | mm | 1 set | ISO 2768-mK | | 1 / 6 |
| SOURCE | | | | DATE | BREAK SHARP EDGES |
| cad/lever_guard.py (CadQuery), drawings.py | | | | 2026-10-23 | 0.2 - 0.5 |




**HALF SECTION A-A**




**VIEW ON SERRATION (from outboard)**

**NOTES**

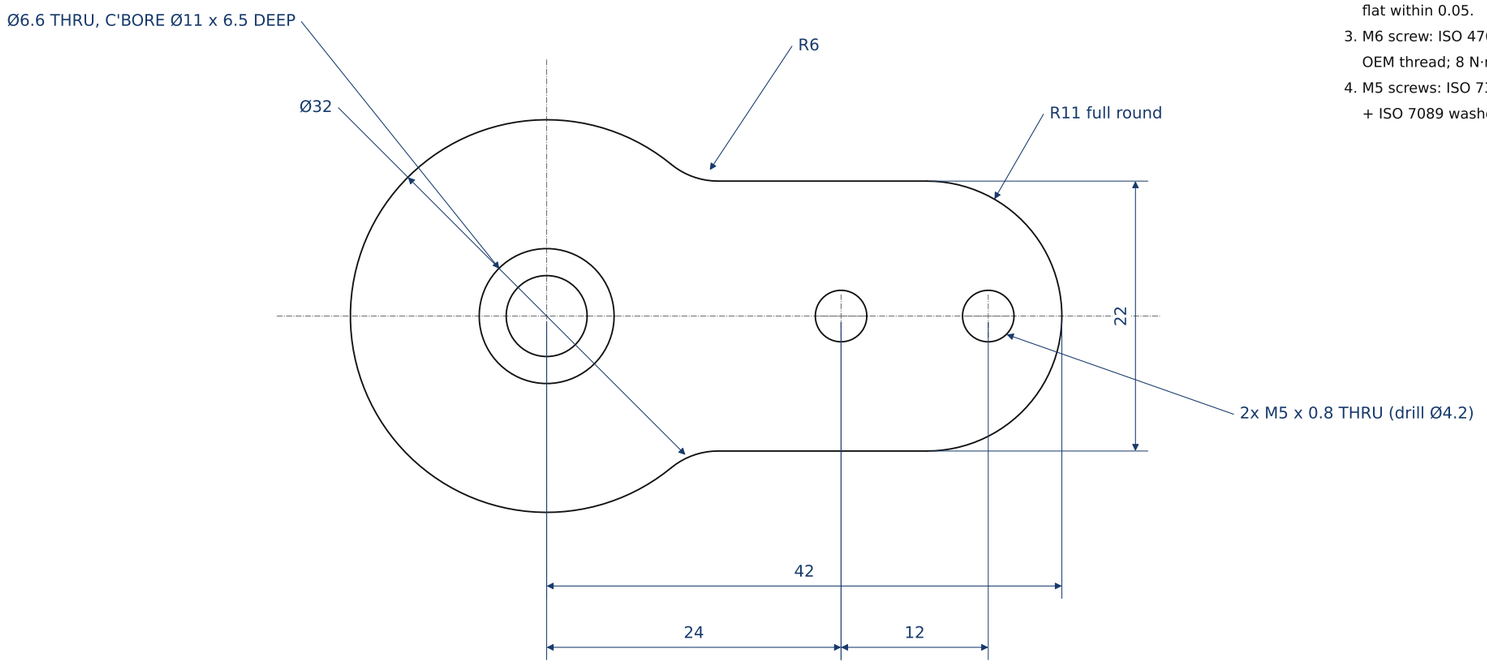
1. Hirth face serration: 36 teeth, 90° included, crests and roots radial through the axis, pitch plane at 12 from the bar-end face, toothed from Ø15 to Ø32: tooth height 1.4 at Ø32. Mates with LG-02 (offset half a tooth).
2. Variant LG-01b: same part without the spigot, for bars with a threaded OEM bar-end insert.
3. Spigot Ø12.4 centres in any bar with ID ≥ 12.5; the expander (LG-04/05) sits behind it.
4. Hard anodise after machining; mask nothing. Serration may be cut with a 90° V-cutter.

RACING LEVER GUARD / UNIVERSAL BAR-END FIT

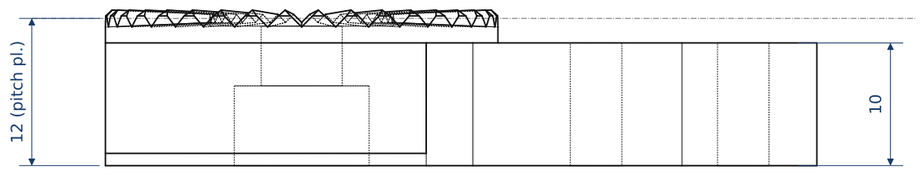
# BAR-END ADAPTER

**LG-01**

| MATERIAL | | | FINISH | | |
|---|---|---|---|---|---|
| EN AW-6061-T6 | | | Hard anodise, colour to order | | |
| SCALE | UNITS | QTY / SIDE | GENERAL TOLERANCES | PROJECTION | SHEET |
| 3 : 1 | mm | 1 | ISO 2768-mK | | 2 / 6 |
| SOURCE | | | DATE | | BREAK SHARP EDGES |
| cad/lever_guard.py (CadQuery), drawings.py | | | 2026-10-23 | | 0.2 - 0.5 |




**OUTBOARD FACE**




**PLAN (lever plane), serration toward the bar**

**NOTES**

1. Hirth serration as LG-01: 36 teeth, 90°, pitch plane on the hub's inner face. Sits half a tooth out of phase with LG-01 so crests meet roots: 10° indexing.
2. Neck edges R1. Outboard face is the blade seat: flat within 0.05.
3. M6 screw: ISO 4762 M6 x 45 (expander) or to suit the OEM thread; 8 N·m, medium threadlocker.
4. M5 screws: ISO 7380 M5 x 14 (x 16 with LG-03b) + ISO 7089 washers, 4 N·m.

RACING LEVER GUARD / UNIVERSAL BAR-END FIT

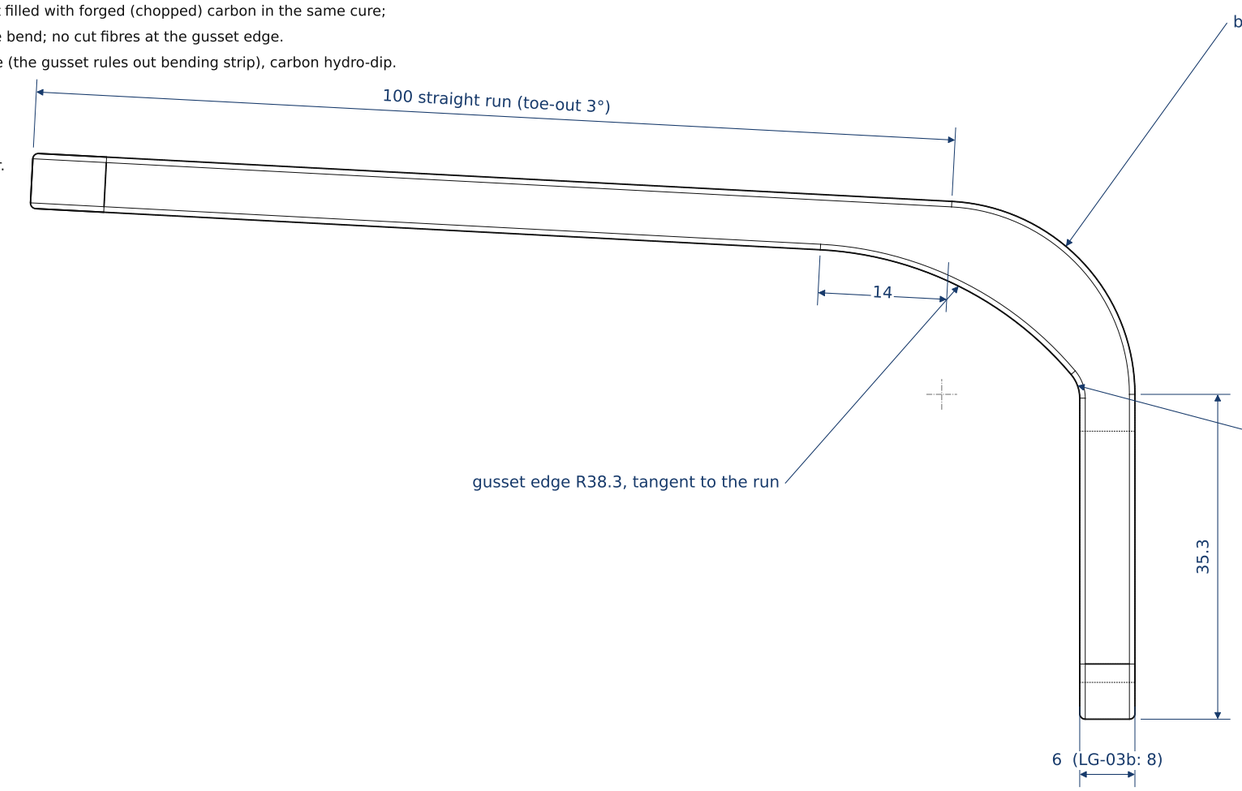
# GUARD ARM

**LG-02**

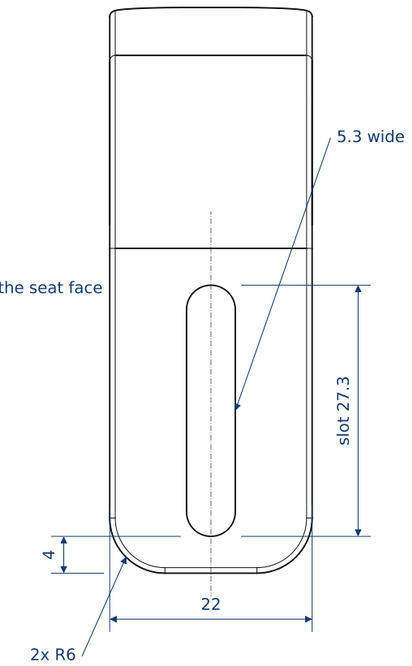
| MATERIAL | | | | FINISH | |
|---|---|---|---|---|---|
| EN AW-6061-T6 | | | | Hard anodise, colour to order | |
| SCALE | UNITS | QTY / SIDE | GENERAL TOLERANCES | PROJECTION | SHEET |
| 2 : 1 | mm | 1 | ISO 2768-mK | | 3 / 6 |
| SOURCE | | | | DATE | BREAK SHARP EDGES |
| cad/lever_guard.py (CadQuery), drawings.py | | | | 2026-10-23 | 0.2 - 0.5 |

**NOTES**

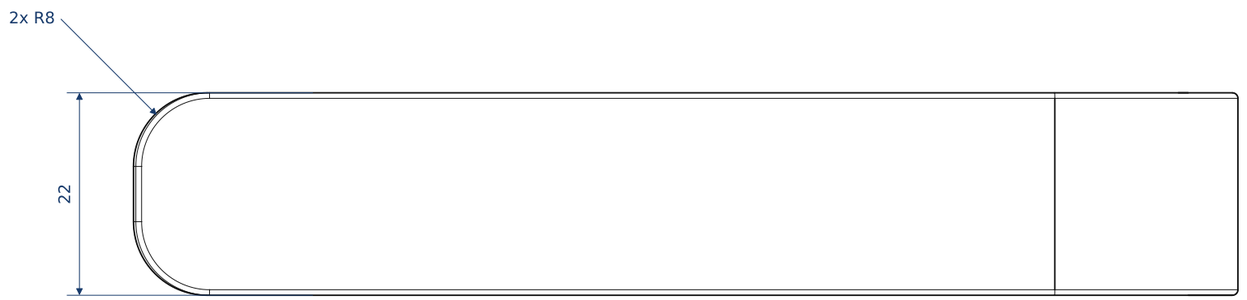
1. Centre-line length 162.6 mm: 35.3 straight + 87° bend on R18 + 100 straight.
2. Gusset inside the bend: section 6 -> 9.7 mm across the bend, about 2.6x the bending strength where a fall loads the blade.
   Its edge is tangent all along (blend into the seat face, arc, run-out on the run): no corner to start a crack.
3. LG-03 carbon: skin 6 ± 0.2 moulded over a male tool, gusset filled with forged (chopped) carbon in the same cure;
   CNC-trim the outline and slot. Lay plies continuous round the bend; no cut fibres at the gusset edge.
4. LG-03 aluminium option: machine from 6061-T6 25 mm plate (the gusset rules out bending strip), carbon hydro-dip.
5. LG-03b 3D print: same outline, 8 thick,
   PA12-CF / PETG-CF, lying flat (lever plane on the bed),
   6 walls, 50 % gyroid; hydro-dip or wrap for the carbon finish.
6. Slot lets the blade slide ±5 mm toward / away from the lever.




**PLAN (lever plane)**




**REAR PLATE (outboard)**



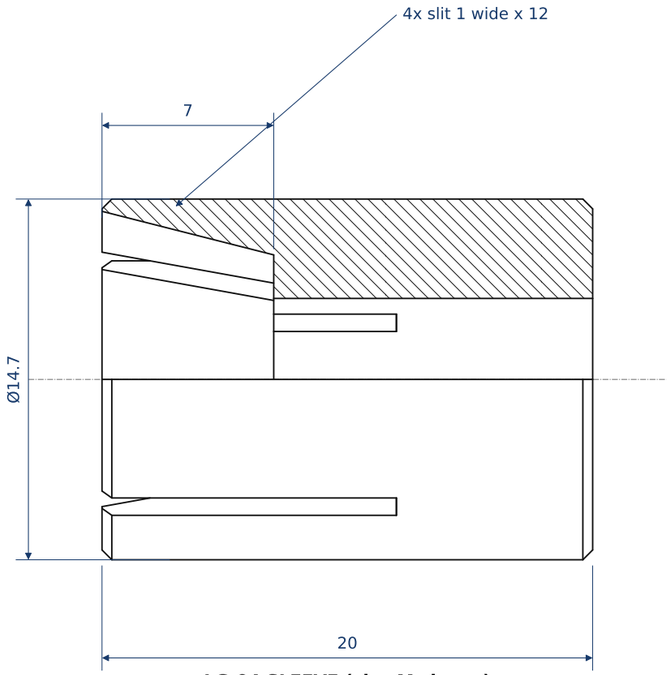

**FRONT VIEW (outer face)**

RACING LEVER GUARD / UNIVERSAL BAR-END FIT

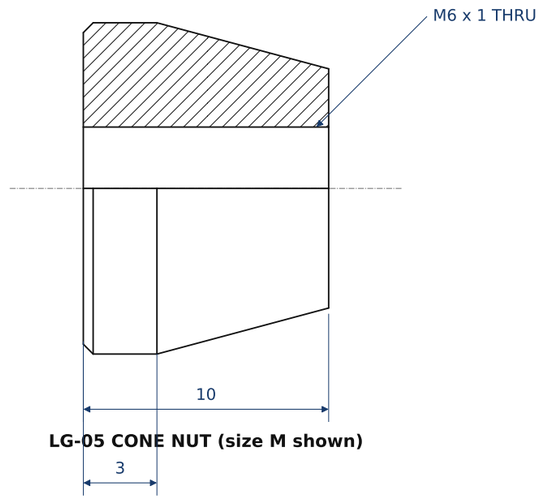
# GUARD BLADE WITH BEND GUSSET

**LG-03**

| | | | | | |
|---|---|---|---|---|---|
| MATERIAL: 3K 2x2 twill carbon / epoxy, forged-carbon gusset | | | | FINISH: Gloss clear coat, edges R0.6 | |
| SCALE: 3 : 2 | UNITS: mm | QTY / SIDE: 1 | GENERAL TOLERANCES: ISO 2768-mK | PROJECTION | SHEET: 4 / 6 |
| SOURCE: cad/lever_guard.py (CadQuery), drawings.py | | | | DATE: 2026-10-23 | BREAK SHARP EDGES: 0.2 - 0.5 |




**LG-04 SLEEVE (size M shown)**




**LG-05 CONE NUT (size M shown)**

**SIZES (all lengths as drawn: sleeve 20, recess 7, cone land 3)**

| Size | Bar ID | Sleeve OD | Recess Ø face / bottom | Cone Ø big / small | Taper (half) |
|---|---|---|---|---|---|
| S | 13 - 15 | 12.7 | 11.7 / 8.4 | 11.5 / 8 | 14° |
| M | 15 - 17 | 14.7 | 13.7 / 10.15 | 13.5 / 9.75 | 15° |
| L | 17 - 19 | 16.7 | 15.7 / 12.15 | 15.5 / 11.75 | 15° |

**NOTES**

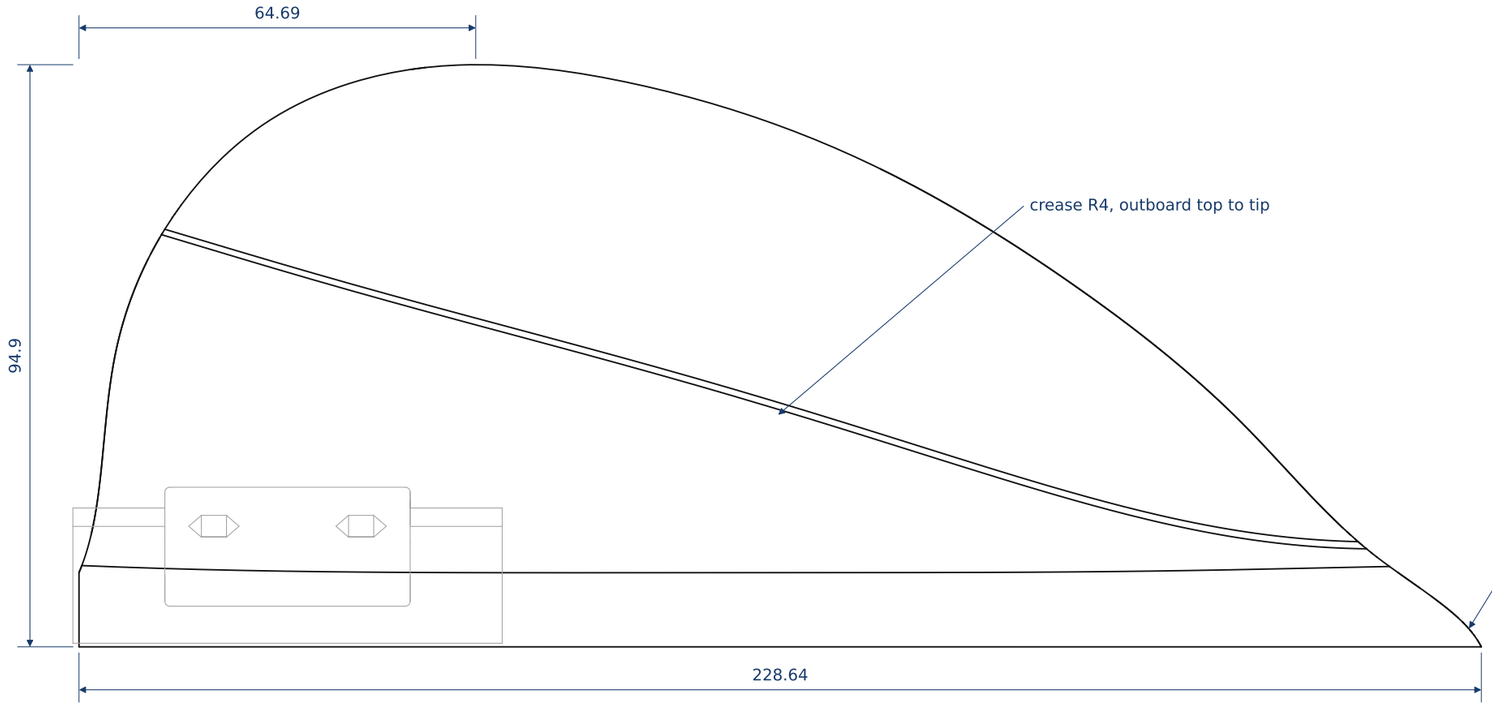
1. Bolt chain: M6 x 45 (ISO 4762) through LG-02 and LG-01 into the cone nut: 8.5 mm thread engagement at fit-up.
2. Tightening pulls the cone into the slit sleeve, which grips the bar bore. 8 N·m. Hold the arm square on its serration while tightening.
3. Pick the size from the bar's measured inside diameter.
4. Sleeve can be printed in TPU 95A (100 % infill) or cut from NBR tube; cone may be steel for higher load.

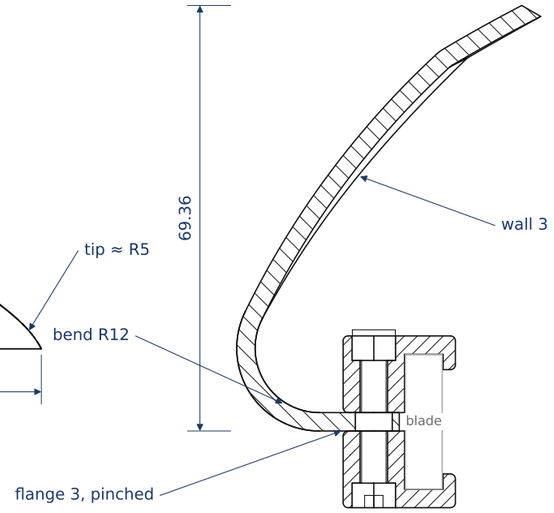
RACING LEVER GUARD / UNIVERSAL BAR-END FIT

# EXPANDER SET

**LG-04 / LG-05**

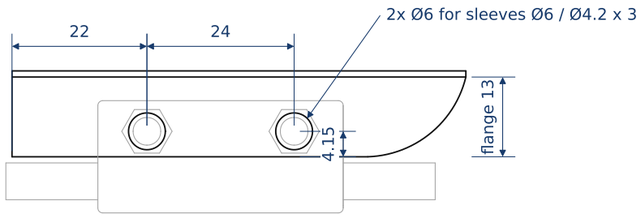
| | | | | | |
|---|---|---|---|---|---|
| MATERIAL: LG-04 TPU 95A or NBR 70A; LG-05 6061-T6 | | | | FINISH: LG-05 clear anodise or zinc | |
| SCALE: 4 : 1 | UNITS: mm | QTY / SIDE: 1 sleeve + 1 cone | GENERAL TOLERANCES: ISO 2768-mK | PROJECTION | SHEET: 5 / 6 |
| SOURCE: cad/lever_guard.py (CadQuery), drawings.py | | | | DATE: 2026-10-23 | BREAK SHARP EDGES: 0.2 - 0.5 |



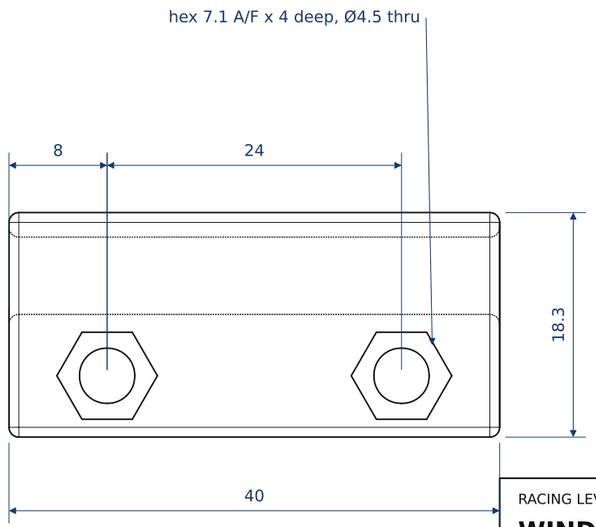




**SECTION THROUGH A SCREW (1:1)**

**LG-06 SHELL (RH), SEEN FROM THE FRONT OF THE BIKE - clamp and blade phantom**




**FLANGE FROM BELOW (pinched by the clamp)**




**LG-07 CLAMP HALF, outer face (2:1)**

**NOTES**

1. LG-06 surface as the STEP files 06 (RH) / 06b (LH, mirror image). Injection-moulded HDPE or PP, or printed ASA / PA12 (flange on the bed, tree supports under the bend and the top). Edges R1.
2. Press the two aluminium sleeves into the flange holes: the clamp load goes through them, not the plastic.
3. LG-07 is one part used twice: the lower half is the upper turned over.
4. Fit with the guard on the bike: halves above and below the blade run, shell flange between their front blocks, 2x M4 x 25 ISO 4762 from below, DIN 985 nyloc nuts captive in the upper half, 2.5 N·m.
5. Loosen to slide: 15-60 mm from the bend. No holes in the blade.
6. Keep the blade slot at -4 or more with the shell fitted (>= 10 mm to the reference lever). Check steering lock, mirrors, reservoir.

RACING LEVER GUARD / UNIVERSAL BAR-END FIT

# WIND GUARD SHELL + SLIDING CLAMP

**LG-06 / LG-07**

| MATERIAL | | | | FINISH | |
|---|---|---|---|---|---|
| LG-06 HDPE or PP, moulded (ASA / PA12 printed); LG-07 EN AW-6061-T6 | | | | LG-06 colour to order, edges R1; LG-07 hard anodise | |
| SCALE | UNITS | QTY / SIDE | GENERAL TOLERANCES | PROJECTION | SHEET |
| 1:1 / 2:1 | mm | 1 pair + 2 halves | ISO 2768-mK | | 6 / 6 |
| SOURCE | | | | DATE | BREAK SHARP EDGES |
| cad/lever_guard.py (CadQuery), drawings.py | | | | 2026-10-23 | 0.2 - 0.5 |